How-To Library

Informatica

Configuring PowerExchange for Teradata Parallel Transporter API to Run Teradata Mappings on the Hive Engine

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Abstract

When you run Teradata mappings on the Hive engine, you can use Teradata Connector for Hadoop to improve performance. This article describes how to configure Teradata Connector for Hadoop to run mappings to read data from or write data to Teradata on the Hive engine.

Supported Versions

- PowerExchange for Teradata Parallel Transporter API 10.2 or later

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Overview

When you run Teradata mappings on the Hive engine, use the Teradata Connector for Hadoop (TDCH) Command Line Edition to increase performance. PowerExchange for Teradata Parallel Transporter API integrates Informatica and TDCH to run Teradata mappings on the Hive engine.

TDCH is a set of API and tools that Teradata Corporation provides for parallel processing of data between Teradata databases and the Hadoop ecosystem of products. You can download TDCH from the Teradata Developer Exchange website.

When you run a Teradata mapping on the Hive engine, the Data Integration Service pushes the mapping to a Hadoop cluster and processes the mapping with one mapper task. When you enable TDCH to run a Teradata mapping on the Hive engine, TDCH uses multiple mapper tasks to read and write the data, which significantly increases the performance.

This article describes how to configure Teradata Connector for Hadoop to run Teradata mappings on the Hive engine. It also provides a list of parameters that you need to define in the TDCH configuration file to read data from or write data to Teradata.

Configuration Tasks

TDCH uses a configuration file named `InfaTDCHConfig.txt` to read and write data. You can define configuration options in the `InfaTDCHConfig.txt` file to specify how you want TDCH to read and write the data.

To enable TDCH to run a Teradata mapping on the Hive engine, perform the following tasks:

1. Complete prerequisites.
2. Install and configure TDCH.

3. Configure the `InfaTDCHConfig.txt` file.
4. Configure the EnableTdch custom property.
5. Specify the Job Monitoring URL in the Hadoop connection properties.

Prerequisites

Before you install and configure TDCH, perform the following tasks:

1. Verify that you have installed and configured the Informatica services and clients.
2. Create a user account in the Teradata Developer Exchange website so that you can download the TDCH installation package from the website.
3. To read data by using TDCH, verify that the following conditions are met:
   - The cluster contains the following folder in the HDFS file system: `/user/yarn/`

     Note: If the folder does not exist, you must create it.
   - If the owner of the `/user/yarn/` folder is not yarn, the user who runs the mapping has rwxrwxrwx permissions on the `/user/yarn/` folder.
4. Ensure that the Hive Client is configured on all the nodes of the Hadoop cluster.
   To verify, run the `hive` command from each node.
5. If the Hadoop cluster is enabled for Kerberos, ensure that the Data Integration Service user has a valid Kerberos ticket on all the nodes of the cluster. If the ticket is not valid, the mapping fails. To generate a valid ticket, run the `kinit` command on each of the nodes of the cluster.

Installing and Configuring TDCH

Install and configure TDCH to run Teradata mappings on the Hive engine.

1. Go to the Teradata Developer Exchange website: http://developer.teradata.com/
2. Download the Teradata Connector for Hadoop Command Line Edition installation package that is relevant for the Hadoop distribution that you use. For example, `teradata-connector-1.4.3-hadoop2.x.noarch.rpm`.
3. Place the `teradata-connector.rpm` on all the nodes of the Hadoop cluster, and install the TDCH installation package.

   The rpm package installs TDCH into the following directory: `/usr/lib/tdch/<version number folder>`
4. Access the following directory on the Informatica server:

   `<Informatica installation directory>\services\shared\hadoop\<Hadoop distribution name>\infaConf\`
5. Find the file named `hadoopEnv.properties`.
6. Perform the following steps in the `hadoopEnv.properties` file:
   a. Set the HADOOP_CLASSPATH to include the paths of the TDCH .jar files and the `hive-site.xml` file as follows:

      `infapdo.env.entry.hadoop_classpath=HADOOP_CLASSPATH=<Path of the dependent TDCH .jar files and the directory where the hive-site.xml file is stored>`
      For example: `infapdo.env.entry.hadoop_classpath=HADOOP_CLASSPATH=/opt/cloudera/parcels/CDH/lib/hive/conf:/opt/cloudera/parcels/CDH/lib/hive/lib/*:/usr/lib/tdch/1.5/lib/*:$HADOOP_CLASSPATH`

b. Set the INFA_MAPRED_CLASSPATH variable to include the paths of the TDCH .jar files and the `hive-site.xml` file as follows:

```
infapdo.env.entry.mapred_classpath=INFA_MAPRED_CLASSPATH=<Path of the dependent TDCH .jar files and the directory where the hive-site.xml file is stored>
```

Configuring the InfaTDCHConfig.txt File

Configure the `InfaTDCHConfig.txt` file to define the parameters that TDCH must use to run Teradata mappings on the Hive engine. Informatica ships a sample `InfaTDCHConfig.txt` file. Customize the sample file based on your requirements.

1. Access the following directory:

   ```
   <Informatica installation directory>\services\shared\hadoop\<Hadoop distribution directory>\infaConf
   ```

2. Find the file named `InfaTDCHConfig.txt`.

   Back up the file before you modify it.

   Note: If the `InfaTDCHConfig.txt` file is missing from the `<Informatica installation directory>\services\shared\hadoop\biginsights_3.0.0.1\infaConf` directory, you must manually copy the file from another Hadoop distribution directory.

3. Use a text editor to open the file and edit the properties.
4. Set the libjars parameter to the paths that contain the dependent .jar files required for TDCH. Perform the following tasks:

   a. To get the list of dependent .jar files required to run a TDCH job, open the readme file from the following location: `/usr/lib/tdch/1.5`

   b. Under **Dependencies**, view the **Hive Job** section to find the list of .jar file names.

   c. Locate the .jar files on the Hadoop nodes in the following directory: `hive/lib`.

      For example: On the Cloudera nodes, verify that the .jar files are available in the following directory:

      ```
      /opt/cloudera/parcels/CDH-5.8.4-1.cdh5.8.4.p0.5/lib/hive/lib/
      ```

      Note: Ensure that all the .jar files are available even though the version of the readme and the .jar file might differ.

   d. Add all the .jar files along with the PATH to the libjars variable in the `InfaTDCHConfig.txt` file.

      For example:

      ```
      libjars=/opt/cloudera/parcels/CDH/lib/hive/lib/libthrift-0.9.3.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/commons-pool-1.5.4.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/commons-dbcp-1.4.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/libfb303-0.9.3.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/jdo-api-3.0.1.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/datanucleus-rdbms-3.2.9.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/datanucleus-core-3.2.10.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/datanucleus-api-jdo-3.2.6.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/antlr-runtime-3.4.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/hive-cli-1.1.0-cdh5.8.4.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/hive-exec-1.1.0-cdh5.8.4.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/hive-jdbc-1.1.0-cdh5.8.4.jar,
      /opt/cloudera/parcels/CDH/lib/hive/lib/hive-metastore-1.1.0-cdh5.8.4.jar
      ```

5. Set other parameters based on how you want TDCH to process the data.
6. Save the file with the name `InfaTDCHConfig.txt`.

Configuration Parameters for Teradata Connector for Hadoop

Configuration parameters determine how TDCH reads data from and writes data to Teradata.

Define the following configuration parameters in the `InfaTDCHConfig.txt` file:

libjars

Defines the paths that contain the TDCH .jar files to which you must set the libjars parameter.

method

Defines the method that TDCH must use to write data to the Teradata database.

You can set the method as batch.insert or internal.fastload.

nummappers

Defines the maximum number of mapper tasks that TDCH must use while writing data.

You can enter an integer value that is greater than or equal to zero. If you enter zero, the number of mapper tasks will be the same as the number of file blocks in HDFS.

Use either this parameter or the numreducers parameter, but not both.

read_nummappers

Defines the maximum number of mapper tasks that TDCH must use while reading data.

You can enter an integer value that is greater than or equal to zero. If you enter zero, the number of mapper tasks will be the same as the number of file blocks in HDFS.

read_method

Defines the method that TDCH must use to read data from the Teradata database.

You can set the read_method parameter to one of the following values:

- split.by.hash
- split.by.value
- split.by.partition
- split.by.amp

For more information about the configuration options, see the Teradata Connector for Hadoop documentation.

Sample TDCH Configuration File

The following `InfaTDCHConfig.txt` file shows sample configuration parameters:

```
#######InfaTDCHConfig.txt
#This file should contain the TDCH attributes specified as "name=value" pairs
#Users should uncomment the required attribute and supply a valid value to it.
#Please refer to the TDCH documentation for description of each attributes
#########

#The attributes such as url, username, password, jobtype, fileformat, separator,
#sourcepaths, sourcedatabase, sourcetable, sourcetableschema, sourcefieldnames,targettable,
#targetfieldnames are taken from the mapping. These attributes if specified in this file
#will be ignored.

#TDCH Export Attributes (for Reader/Writer)
#This "libjars" attribute is mandatory. Please uncomment it, based on Hive or Blaze mode run.
#If not specified the mapping will fail to run. The dependent
#jar files are specified below. Replace '<HIVE_HOME>' with path to hive installation directory
#and replace 'version' with actual version number of the jar files.  Please refer TDCH README
for
#the actual list of dependent jars.
```

```
#for Hive only
#libjars=<HIVE_HOME>/lib/libthrift-version.jar,<HIVE_HOME>/lib/commons-pool-version.jar,
<HIVE_HOME>/lib/commons-dbcp-version.jar,<HIVE_HOME>/lib/libfb303-version.jar,
<HIVE_HOME>/lib/jdo-api-version.jar,<HIVE_HOME>/lib/datanucleus-rdbms-version.jar,
<HIVE_HOME>/lib/datanucleus-core-version.jar,<HIVE_HOME>/lib/datanucleus-api-jdo-version.jar,
<HIVE_HOME>/lib/antlr-runtime-version.jar,<HIVE_HOME>/lib/hive-cli-version.jar,
<HIVE_HOME>/lib/hive-exec-version.jar,<HIVE_HOME>/lib/hive-jdbc-version.jar,
<HIVE_HOME>/lib/hive-metastore-version.jar

#for Blaze only
#Give absolute path of these two jars - avro-mapred-version-hadoop2.jar,
avro-version.jar present in hadoop distribution.
Hadoop distribution may have multiple versions of these two jars,
Its recommended to use latest version above 1.7.4 present in hadoop distribution.
Location of these jars may differ based on hadoop distribution. e.g.
#libjars=/opt/cloudera/parcels/CDH/lib/avro/avro-version.jar,
/opt/cloudera/parcels/CDH/lib/avro/avro-mapred-version-hadoop2.jar

#This "tdchjar" attribute is optional for Hive but mandatory for Blaze. Give absolute path to
tdch jar file.
#If the jar path specified then it will be used in constructing the TDCH command line.
#example tdchjar=/usr/lib/tdch/version/lib/teradata-connector-version.jar
#tdchjar=

#This attribute is optional. The method Teradata Connector for Hadoop uses to export data
#to Teradata System. Supported Values are batch.insert, multiple.fastload, and
#internal.fastload. batch.insert is the default value.
#method=

#This attribute is optional. The method Teradata Connector for Hadoop uses to import data
#to Teradata System. Supported Values are split.by.hash, split.by.value, split.by.partition and
split.by.amp
#split.by.hash is the default value.
#read_method=

#This attribute is optional. The maximum number of output mapper tasks.
#If the value is zero, then the number of mappers will be same as the number of file
#blocks in HDFS. Use either this parameter or 'numreducers', but not both. Set nummappers for
writer and set read_nummappers for reader.
#nummappers=
#read_nummappers=

#This attribute is optional. The database with which Teradata Connector for Hadoop uses to
#create staging tables.#stagedatabase=

#This attribute is optional. The name in Teradata system with which Teradata Connector for
#Hadoop uses to create staging table, if staging is required. Its length cannot exceed 20
#characters. It should be used when the target Teradata table has name exceeding 20
#characters. The stagetablename should be less than 20 characters.
#stagetablename=

#This attribute is applicable for only Blaze Mode(Not for Hive Mode).
#It is Base path for staging temporary avro data file and schema file at
cluster.#tdchstagingdir=
```

Configure the EnableTdch Custom Property

You must configure the EnableTdch custom property to enable TDCH to run a Teradata mapping on the Hive engine.

Set the custom property name as EnableTdch and the value as True for the Data Integration Server. After you set the property, recycle the Data Integration Server.

Specifying the Job Monitoring URL in the Hadoop Connection Properties

You must specify the Job Monitoring URL in the Hadoop connection properties to enable TDCH to run a Teradata mapping on the Hive engine.

1. In the Developer tool, click **Window** > **Preferences**.
2. Select **Informatica** > **Connections**.
3. Expand the domain.
4. Select the connection type **Hadoop** in the **Available Connections** list.
5. On the **Hive Pushdown** tab, specify the **Job Monitoring URL** value. Perform one of the following tasks:
   - On the **Common Attributes** tab, copy the value of the **Resource Manager Address** field and paste the value in the **Job Monitoring URL** field available on the **Hive Pushdown** tab.
   - If the **Resource Manager Address** field is blank, specify the **Job Monitoring URL** in the following format:

   <hostname>:<port>, where <hostname> is the host name or IP address of the JobHistory server and <port> is the port on which the JobHistory server listens for remote procedure calls (RPC).

   For example, enter: myhostname:8021

   You can get the value for the Job Monitoring URL from mapred-site.xml. The Job Monitoring URL appears as the following property in mapred-site.xml:

   ```
   <property>
   <name>mapred.job.tracker</name>
   <value>myhostname:8021 </value>
   <description>The host and port that the MapReduce job tracker runs at.</description>
   </property>
   ```
6. Click **OK**.

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